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Private ownership of water and wastewater systems: Assessing health impacts

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ABSTRACT

This study examines the impact of private ownership of water and wastewater systems on disease reduction linked to sanitation in Brazil from 1998 to 2021. It updates Saiani and de Azevedo (2018), which analyzed the period 1995–2008, by incorporating over a decade of additional data, key policy changes such as the 2020 Sanitation Law, and employing the Callaway–Sant'Anna Staggered DID methodology to address heterogeneity in treatment effects. Our findings reveal mixed results: while some municipalities achieved reductions in morbidity rates, others showed no change or increases, underscoring the context-dependent nature of privatization outcomes. A notable example is the case of Tocantins, where transitioning from a hybrid private-state model to full private ownership led to a significant decrease in disease morbidity, particularly among the most affected age groups. These advancements provide a robust, updated perspective on the privatization debate, offering valuable implications for policy and practice.

1. Introduction

Globally, most diarrheal diseases, responsible for nearly two million deaths a year, can be attributed to unsafe water supplies, as well as inadequate sanitation and hygiene. Most of the deaths are among children in developing nations (Charrois and Jeffrey, 2010).

The quality of sanitation services plays is essential in reducing the prevalence of diarrheal diseases, particularly among vulnerable populations such as children under five years of age in low and middle-income countries. Access to improved sanitation facilities, including ventilated pit latrines and toilets connected to a public sewer or septic system, significantly reduces the risk of diarrhea. The study by Merid et al. (2023) highlights that improved sanitation alone is associated with a 16% reduction in the risk of diarrheal disease, while combining improved access to water and sanitation leads to a reduction of 24.5%. The synergistic effect underscores the importance of improving infrastructure and service quality to address public health challenges effectively.

This perspective leads us to a worthy question. Is the privatization of sanitation services conducive to improving public health? Privatization can change the incentives and thus affect how well services are delivered. Since epidemiological indicators are directly related to the standard of sanitation services, privatization could influence public health.

In 2020, a notable development was observed in Brazil's water and wastewater sector when the Brazilian government approved a new law that could significantly increase the private sector's participation in the field. This legislative change adds a fresh layer of relevance to the ongoing discussion regarding the effects of privatization in this essential services domain.

Saiani and de Azevedo (2018) noted that in Brazil, the responsibility for water supply and wastewater collection lies with municipalities, and they had the flexibility to choose four distinct organizational modes to provide these essential services: public at the state level, public local, private local and private with state rights.

When Brazil privatized part of its sanitation services between the late 1990s and early 2000s, it provided an opportunity to study how private companies can affect public health. From 1995 to 2008, many of Brazil's towns switched their water and wastewater services to private firms. Different municipalities decided to change their sanitation service providers at different times and in different ways, allowing substantial data to be collected and studied by Saiani and de Azevedo (2018).

However, while private companies operated under the same overarching rules (Estrin et al., 2009), the regulatory environment in Brazil is far from uniform. As highlighted by Carvalho et al. (2023), variations in regulatory incentives, governance structures, and enforcement intensity significantly influenced the performance of water and sewage companies. These differences suggest that the effects observed from privatization might also reflect the heterogeneity in regulatory practices

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across municipalities. This article provides valuable insights into which incentives work best, as discussed by Carvalho et al. (2023).

All the new data available, presents an opportunity to revisit and update the existing findings by Saiani and de Azevedo (2018). The abundance of recent data and new methodologies allows for a more nuanced and contemporary analysis, offering insights into Brazil's evolving landscape of sanitation services.

2. Context

In Brazil, private service provision was operated in two different ways. The first form of operation was termed "local private", involving entirely private companies delivering sanitation services at a local level, typically within smaller populations. In particular, as the Brazilian Ministry of Cities (Robles, 2008) indicated, the local private mode worked relatively homogeneously. All municipalities were obligated to adhere to Federal Concession Law No. 8987, and the contracts share fundamental governance features, allowing them to be treated collectively as a group. However, these contracts were unequal and differed.

The second approach was labeled "private with state rights", where a private business responsible for sanitation services shared decision-making powers with the state government. This mixed ownership model, neither entirely public nor private, adds an intriguing element for study and examination. This organizational model emerged from the privatization of the state sanitation company in Tocantins, a state located in the northern region of Brazil. Tocantins exhibit a per capita income close to the Brazilian median and low population density, with approximately one-fourth of the national average (5.49 inhabitants per square kilometer).

Regarding public provision, Saiani and de Azevedo (2018) highlighted the importance of state sanitation companies (SSC), controlled by state governments and tasked with providing services in multiple municipalities within their respective states. They also emphasized the existence of public providers supervised by individual municipalities or as part of a consortium involving neighboring municipalities. It is important to note that both Mato Grosso and Tocantins, Brazilian states, represented two exceptions to the norm, as every Brazilian state except for these two had a state sanitation company (SSC) controlled by state governments and responsible for providing services to several municipalities in their respective states. In 2008, Mato Grosso extinguished its state sanitation company after a long process that began in 2000, and Tocantins opted to transfer control of its state company to a private operator.

The privatization of Tocantins SSC in 1998 was the only case in Brazil where the state became a minority shareholder (Freitas et al., 2017). Tocantins state remained the owner of golden shares that gave rights to appoint the Director of Planning, one member of the Board of Directors, and veto rights related to changes in the company's statute, such as location and reduction of planned targets. Most municipalities in Tocantins, in different years, opted to delegate sanitation services to this private provider with state rights. There were still some cases of local public providers in Tocantins and a single case of a municipality that opted for a local private provider.

In 1998, 43.2% of the state's population had access to the water supply network, and wastewater collection and treatment services were virtually nonexistent in Tocantins, reaching only a negligible population (Soares et al., 2017). In Palmas, the capital, twelve years after the privatization process, in 2010, more than 70% of the population was connected to the water supply network. However, in that year, only 12 municipalities in the state had wastewater collection services, representing 13.56% of the state's population. Of the 12 municipalities with wastewater services, the private with state rights provider was responsible for nine, while the other three were under the responsibility of another service provider.

Due to the slow progress of sanitation services in small municipalities and rural areas of Tocantins and the difficulties in obtaining federal funds for the private provider with state rights, Saneatins, the state government established an autonomous agency in March 2010 to provide essential sanitation services in the state — Aguatins. This agency aimed to complement the services provided by the Company. In August of the same year, Aguatins and Saneatins signed a Protocol of Intentions, agreeing that the agency would take over sanitation services in the rural areas of all municipalities and in the urban areas of 78 municipalities, while Saneatins would continue operating in the urban areas of 47 municipalities in Tocantins.

The protocol's agreement materialized in 2013, with the agency (which changed its name in 2011 to ATS—Tocantinense Sanitation Agency) assuming the agreed-upon services. Saneatins retained control over 39 of the 50 municipalities with the highest urban population in the state, while ATS took over services in municipalities with a less significant urban population. In December 2011, Odebrecht Ambiental S.A., with FI-FGTS as a partner holding a 30% stake, acquired EMSA's participation in Saneatins (76.5%), becoming the company's majority shareholder. The state government's shares (23.5%) in the Company were transferred to ATS. In mid-2014, Odebrecht Ambiental S.A. acquired the total shares of Saneatins managed by ATS (23.5%), gaining full control of the Company.

In 2016, Brookfield Brazil Capital Partners LLC and the BR Ambiental Investment Fund, managed by Brookfield Asset Management, took over 70% of Odebrecht Ambiental's capital. The FI-FGTS investment fund maintained its 30% stake in the company. Therefore, private companies with state rights operations have no longer been operating since 2014, considering that the state government no longer has any shares in this company (Soares et al., 2017).

The approval of the new sanitation legal framework in 2020 introduced fundamental principles designed to drive the country out of its position among the lowest-ranked nations regarding the coverage of global water and wastewater services for the population. This new legislation delineates a structured path for the sector, underpinned by three pivotal pillars: establishing appropriate regulation, promoting increased competition, and enhancing economies of scale in service provision. These interconnected pillars collectively aspire to universalize essential sanitation services, as noted by ABCON SINDCON, the National Association and Union of Private Concessionaires of Public Water and Wastewater Services (ABCON, 2020).

Published on May 31, 2021, Decree 10.710/21 represents a significant step in the regulation of Law 14.026/20, reinforcing the core principles of the legal framework. This decree establishes the methodology to verify the economic and financial capacity of water supply and wastewater service operators to meet the universalization goals established in the new legal framework for sanitation. All service providers operating under program contracts (public operators) and those selected through bidding processes (private operators) intending to include universalization targets must align themselves with the decree and demonstrate their economic and financial capacity to fund the necessary investments.

In the initial phase, a financial analysis of the company's financial status is conducted based on its financial statements from the past five years. This analysis considers assets, liabilities, revenues, and the entity's capacity for investment, among other factors. According to data from the "Panorama da Participação Privada no Saneamento 2021" (ABCON, 2021), an annual publication by ABCON SINDCON, private concessionaires currently provide full or partial sanitation services, depending on the concession model adopted in each municipality, to 15% of the country's population (approximately 31.6 million people). They are present in 7% of the municipalities.

Taking into account recent auctions of companies such as Casal (AL, 2021), Sanesul (MS, 2020), Cedae (RJ, 2021) and the municipality of Cariacica (ES, 2020), private operators now directly or indirectly serve 17% of the population. The ABCON SINDCON expectation is that

Table 1
Number of privatized companies between 1996 and 2021.
Source: SNIS

Year	Water	Water and Wastewater	Wastewater	Total
1996	0	1	0	1
1997	0	0	0	0
1998	4	5	0	9
1999	1	4	0	5
2000	1	4	0	5
2001	2	3	0	5
2002	1	2	0	3
2003	1	0	0	1
2004	126	9	1	136
2005	1	0	0	1
2006	6	4	0	10
2007	5	0	0	5
2008	10	3	0	13
2009	2	1	0	3
2010	1	4	0	5
2011	2	3	1	6
2012	2	4	1	7
2013	2	2	0	4
2014	4	1	0	5
2015	8	6	0	14
2016	3	3	3	9
2017	1	3	1	5
2018	6	7	0	13
2019	3	1	0	4
2020	6	4	0	10
2021	12	31	0	43
Total	**210**	**105**	**7**	**322**

they should reach at least 40% by 2030. Private sanitation concessions currently account for 33% of the total investment made by companies in the sector. In 2019, private operators invested R$ 4.85 billion, while the total sector investment reached R$ 14.8 billion, which covers all operators (ABCON, 2021) (see Table 1).

In general, sanitation in Brazil has shown slow but steady progress in recent years. In 2022, the total coverage of the water supply reached 84. 9% of the population, while the coverage of the sewer service increased to 56%, marking slight improvements in access. Furthermore, the rate of treated wastewater relative to total generated wastewater rose to 52 2%, indicating gradual progress in addressing sanitation challenges. However, significant gaps remain, as approximately 34 million people still lack access to potable water and more than 101 million people live without access to wastewater collection services. To achieve universalization, providing 99% of the population with potable water and 90% with collected and treated wastewater, a total investment of R$ 753 billion over the next 12 years is required. This amount includes R$ 255 billion allocated to restore depreciated networks and existing assets (ABCON, 2024).

According to data from the Brazilian Institute of Geography and Statistics (Pesquisa Nacional de Saneamento Básico, 2017), Brazil consists of 5568 municipalities, of which a substantial 5548 are endowed with at least one water service provider. In stark contrast, a more modest proportion, precisely 3359 municipalities, has wastewater treatment facilities.

In 2023, Brazil introduced significant changes to its sanitation legal framework. First, municipalities must form regional service blocs by December 31, 2025, integrating smaller municipalities with more developed areas to ensure technical and financial viability. Second, the requirement for competitive contracting was reinforced, restricting no-bid agreements for state-owned companies to exceptional cases where open competition is either legally exempt or proven unfeasible. Third, the deadline for sanitation service providers to demonstrate their economic and financial capacity was set for December 31, 2023, ensuring that only financially stable companies remain in the sector. Lastly, the 25% cap on private sector participation in public–private partnerships (PPPs) was removed, enabling greater private involvement in sanitation projects, according to the Brazilian newspaper (G1, 2023).

3. Literature review

Is privatizing sanitation services conducive to improving public health and broader accessibility to these services? The initial question has been explored in the research of Galiani et al. (2005); this study found that fewer babies died when Argentina made sanitation private. However, Granados and Sánchez (2014) received different results when they studied the case of Colombia; the results obtained indicate that municipalities that allowed private companies to provide sanitation services exhibit a slower reduction of child mortality rates and lower increases in water coverage than those that did not have the involvement of private providers.

Several studies have examined the shift to private ownership, and their findings are mixed. They indicate that local regulations, laws, competition, and the structure of organizations can all influence the outcomes (Sabbioni, 2008; Estrin et al., 2009; Hailu et al., 2012; Ménard and Saleth, 2013; Tan, 2012). Together, these studies suggest it is important to explore the second question: How do different modes of privatizing sanitation services affect health indicators?

Saal et al. (2004), conducted a comprehensive study evaluating the impact of privatization and economic regulation on the water and waterwaste industry in England and Wales. Following the industry's privatization in 1989, a new regulatory framework and a substantial capital investment program were implemented to enhance water and environmental standards. Productivity growth estimates using quality-adjusted output indices indicate that overall total factor productivity growth did not improve post-privatization, despite reductions in labor usage.

In addition, their examination of the total price performance indices reveals a notable trend in which the escalation of out- put prices outpaces the increase in input costs. This observation is identified as a key factor contributing to increased economic profits since privatization.

The common belief is that a private company usually wants to cut costs more than a state-run company because it can pocket the savings from cost cuts. However, a challenge arises when the quality of service cannot be perfectly regulated through contracts, and there is no market mechanism to reward or compensate for maintaining high quality. In such cases, strong incentives for cost reduction in a private operator can lead to a decrease in the quality of the service they provide, as noted by Hart et al. (1997) in a study showing that public provision would be a better solution in the prison system.

Similarly to prisons, consumers of essential sanitation services often lack the option of choosing among suppliers. There is the same issue: There are no market forces that ensure service quality. Thus, essential sanitation shares some common traits with sectors where service quality and costs must be balanced.

Keeping quality standards in concession contracts can be difficult due to a variety of contractual hazards, as Ménard and Saleth (2013) pointed out. Consequently, sanitation services can be seen as a case where a private operator can be incentivized to deliver lower quality levels.

Given that the quality of sanitation services has a direct impact on the prevalence of various diseases, this trade-off can have tangible consequences for the health of individuals in areas with privately provided sanitation services. In this context, the quality of sanitation services becomes a critical factor affecting the population's well-being and health outcomes

There is no substantial dispute on the impact of privatizing sanitation services on costs, a point previously established in the study by Jiang and Zheng (2014) for China and the study by Sabbioni (2008) for Brazil. However, current research delves into the more contentious issue of how privatization affects service quality and its implications for epidemiological metrics associated with water and sewage services.

Saiani and de Azevedo (2018) found that the local private mode of organization decreases morbidity rates in sanitation-related diseases and in the age groups most vulnerable to sanitation interventions.

Table 2
Summary of literature review findings and implications.

Study	Main findings	Implications
Galiani et al. (2005)	Reduced child mortality in Argentina.	Privatization can improve public health.
Granados and Sánchez (2014)	Slower reduction in child mortality in Colombia.	Local context and regulation are key.
Saiani and de Azevedo (2018)	Mixed results for different privatization models.	Impact depends on privatization type.
Deutschmann et al. (2023)	Reduced child diarrhea rates in Senegal.	Productivity improvements aid health.
Chen et al. (2022)	Optimal privatization depends on economics.	Consider governance and efficiency.

Regarding the mode of organization with state rights, there is no robust evidence of deterioration in quality in water and wastewater services but no improvement either. In summary, their findings contradict the prediction of the cost-quality trade-off.

The most recent literature based on previous research (Saiani and de Azevedo, 2018) has mixed results. Chen et al. (2022) present a mixed duopoly model, examining market equilibrium and optimal proportions of state-owned shares in various competition modes. Their findings emphasize the nuanced impact of product differentiation, efficiency gaps, and reform costs on equilibrium results. The study highlights the need for caution in state-owned enterprise strategies and recommends a careful consideration of complex economic factors in determining share proportions. Furthermore, the authors stress that the choice between privatization and nationalization depends on specific conditions with implications for social welfare. The research underscores the importance of assessing influential factors to establish a mode of competition conducive to social well-being.

However, Deutschmann et al. (2023) show that the privatization of wastewater treatment centers in Dakar, Senegal, results in a substantial increase in legal sanitary waste dumping, indicating a 74% average increase or approximately 1640 additional monthly trips to treatment centers. This surge in legal dumping is attributed to enhanced productivity in all trucks, not only those associated with the privatized center. The data from the household survey revealed a reduction of 5% in downstream prices for legal sanitary dumping after privatization. In addition, the data indicate a decrease in diarrhea rates among children under five years of age in Dakar compared to secondary cities, highlighting positive public health outcomes after privatization (see Table 2).

The criteria for including research articles in the literature review focused on selecting works published in relevant and reputable journals.

4. Data

Hospitalization data are derived from the Hospital Information System of Brazil's Unified Health System (SUS), managed by the Brazilian Ministry of Health and accessible through the SUS Department of Information Technology (DATASUS, 2023).

Brazil's public healthcare system, SUS, only grants public or private hospitals subsidies if they issue data on hospitalization authorizations (AIH) to municipal or state managers. After this, DATASUS (2023) processes this information, facilitating the precise calculation of morbidity rates.

It is important to note that individuals may be hospitalized or have their death records documented in a municipality different from their residence. Operating under the assumption that sanitation conditions impact an individual's health in their municipality of residence, hospitalization data is collected based on the place of residence.

Morbidity rates are expressed as the number of individuals admitted to a hospital with the selected diseases, multiplied by 100,000, and divided by the total population to obtain the rates for 100,000 inhabitants. The Brazilian Institute of Geography and Statistics (IBGE) provides yearly municipal population data.

Data on sanitation service provision methods are sourced from the Brazilian Ministry of Cities, as mentioned in SNIS (2023). Previous research by Saiani and de Azevedo (2018), which analyzed data collected between 1994 and 2008, identified 70 municipalities that transitioned from public to local private providers. Additionally, 123 municipalities switched from public providers to private providers with state rights, a unique organizational form in Tocantins. This structure retains a golden share and grants veto powers to the state.

This study aims to evaluate the impact over different periods and to identify potential differences between different time periods. Morbidity data from 1998 to 2023 is available in DATASUS (2023), while information on sanitation services in municipalities can be accessed from 1995 to 2021. A municipality is considered to have privatized its sanitation system if it opted for a private provider for water, waterwaste, or both between 1998 and 2021, and continued with that private system until the most recent records available. The diseases included in the study are listed below.

4.1. Disease selection

Diarrheal diseases: Amebiasis, Cholera, Diarrhea, and gastroenteritis, all presumed to be of infectious origin, and other intestinal infectious diseases.

Non-diarrheal diseases: Conjunctivitis and other conjunctival disorders, Schistosomiasis, Yellow fever, Dengue [classic dengue], Unspecified malaria, Unspecified leishmaniasis, Filariasis, Typhoid and paratyphoid fevers, Unspecified leptospirosis, Trachoma, Mycoses, Trypanosomiasis.

Diseases not related to sanitation: Other forms of tuberculosis, Other forms of tetanus, Other forms of syphilis, Measles, Rubella, Obesity, Hemorrhoids, Influenza [flu], Osteoarthritis.

The selection of diseases for all disease groups was guided by the previous findings of Wennemo and Irene (1993) and Feachem et al. (1983).

5. Methodology

To evaluate the impact of privatization on the quality of sanitation services, we use the empirical methodology proposed by Galiani et al. (2005) and later refined by Saiani and de Azevedo (2018). Epidemiological research indicates that improved sanitation services, encompassing access and quality improvements, directly influence various diseases, particularly those associated with water (Cairncross and Feachem, 2018; Mara and Feachem, 1999).

As sanitation services are linked to diseases in the water and sewer domains, it does not matter whether a municipality privatizes its water, wastewater, or water and sewer services. The classification of a city as having privatized its sanitation is valid regardless of whether it involves individual services or a combination of both. Therefore, the presence of a private company that oversees any aspect of water or

Table 3
Diseases by category and Group.

Category	Group	Diseases
Faecal-Oral	Diarrheic	Cholera, Salmonella, Amebiasis, Isosporiasis, other intestinal infections
Insect vector	Enteric fever	Typhus, Paratyphoid Fevers
	Others	Hepatitis A, Polio, Leptospirosis, Ascaridiasis, Tricuríase, Lymphatic Filariasis, Malaria, Chagas disease, Dengue, Yellow fever, Leishmaniasis
	Close to water	Sleeping sickness
Contact with water	Skin penetration	Schistosomiasis
	Ingestion	Helminth infections, tapeworm, cysticercosis
Hygiene related	Eye diseases	Trachoma, Conjunctivitis
	Skin diseases	Dermatophytoses, superficial mycoses

Table 4
Control measures by category.

Category	Control measures
Faecal-Oral	Water supply, water quality, wastewater collection, health education, housing improvements, rodent control
Insect vector	Water quality, wastewater, drainage of rainwater, health education, elimination of procreation sites, biological control, mosquito nets
Contact with water	Water quality, wastewater, reduction of contaminated waters, health education, control of animal reservoirs, meat inspection
Hygiene related	Water supply, water quality, wastewater, health education

wastewater operations within a municipality warrants a classification of 1. On the other hand, in the absence of a private operator in a city, the classification is denoted as 0.

The morbidity of children under the age of five can be categorized into three groups based on the age of occurrence: neonatal (from birth to the 27th day), postnatal or late infant (from the 28th day to one year old), and child (one to five years old). Neonatal is primarily attributed to complications during pregnancy or delivery, such as prematurity, congenital malformations, and low birth weight (Victora et al., 1994). Sanitation conditions can affect neonatal mortality by contributing to maternal morbidity throughout pregnancy. Post-neonatal is predominantly associated with the complex of diseases known as the diarrhea-pneumonia-malnutrition complex, also influenced by sanitation quality (Wennemo and Irene, 1993; Wang and Limin, 2003). Combined, neonatal and postnatal morbidity is commonly referred to as infant morbidity, representing most child diseases within the first year.

In the age group of one to five years, the cases tend to decrease compared to the first year. Children in this age range have a more developed immune system and the least healthy individuals often experience diseases during the first year of life. Consequently, children aged one to five years can experience illnesses related to sanitation conditions, but the risk of mortality is reduced (Wennemo and Irene, 1993; Wang and Limin, 2003; Victora et al., 1994).

This study takes advantage of our dataset's temporal and geographical variations, which enables us to estimate the impact of several forms of private operation compared to public provision, focusing on morbidity rates. Our identification strategy relies on information from the epidemiological literature, linking sanitation interventions to specific types of disease (sanitation-related diseases and others) and particular age groups.

Sanitation-related diseases are classified according to pathogen life cycles and transmission routes. Each category encompasses specific disease groups highly susceptible to sanitation quality. This study includes corresponding control measures for each disease group, incorporating water quality and sanitation collection.

As discussed above, the influence of sanitation services on health is primarily observed in early childhood. Vulnerability during these early years is due to the developing immune systems of children, increased water and food consumption compared to body weight, and behaviors such as hand-to-mouth contact, all of which amplify the risk of contamination due to water and wastewater quality.

Both Tables 3 and 4 are based on the classifications and frameworks presented in Wennemo and Irene (1993) and Feachem et al. (1983). These studies comprehensively analyze sanitation-related diseases and appropriate control measures to mitigate their impact. Wennemo and Irene (1993) focuses on the relationship between sanitation conditions and infant morbidity, while (Feachem et al., 1983) offer detailed categorizations of waterborne and hygiene-related diseases. Together, these foundational works inform the categorization of diseases and the corresponding public health interventions summarized in the tables.

Our goal is to examine the established impact of sanitation on specific diseases and age groups to understand the influence of various methods of providing sanitation services on overall quality. Recognizing that mortality and morbidity rates may be influenced by factors unrelated to sanitation services, we extend our analysis to include these indicators in diseases unrelated to sanitation across diverse age groups.

Suppose privatization leads to a decline in sanitation quality. In that case, we anticipate a more pronounced impact on morbidity rates related to sanitation-related diseases in children aged 0 to 5 years, with a lesser or negligible effect on other diseases and age groups.

In contrast to the focus of Galiani et al. (2005) and following what Saiani and de Azevedo (2018) developed, our emphasis is on morbidity rather than the mortality rate. The literature underscores the importance of morbidity, mainly due to diarrheal diseases, as a robust epidemiological indicator to assess the immediate and direct impacts of sanitation interventions on health (Esrey and Habicht, 1987; Feachem et al., 1983; Esrey et al., 1985; Briscoe et al., 1986).

Morbidity proves to be a more reliable metric than mortality because diseases related to sanitation, such as diarrhea, directly lead to hospitalization and indirectly contribute to opportunistic illnesses that may eventually be recorded as the cause of death. For example, complications from diarrheal diseases, such as pneumonia, can arise (Scrimshaw et al., 1968; Saad and Murad, 1968). Consequently, the morbidity rate emerges as a more reliable epidemiological indicator to evaluate the quality of sanitation services.

First, it is necessary to identify a control group of municipalities exhibiting similar observed characteristics to those in the treatment group. Propensity score matching (PSM) is used for this purpose, generating weights to align the characteristics of the two groups and establish common support, as described by Caliendo and Kopeinig (2008).

We used the nearest neighbor approach, considering a simple nearest neighbor classifier. Then, the logistic regression component assigns weights to the nearest neighbors based on their distance and predicts the probability using these weighted neighbors.

$$h(x_{\text{test}}) = \frac{1}{1 + e^{-\theta^T w x_{\text{nearest}}}}$$

- $h(x_{\text{test}})$ is the predicted probability that $y = 1$ for the test instance x_{test}.
- w is a weight assigned to the nearest neighbor based on its distance.
- x_{nearest} is the feature vector of the nearest neighbor.
- θ is the parameter vector learned through logistic regression.

Table 5
Independent variables (Explanatory variables X_i).

Variable	Description
Piped water	Ratio of households with water supply to total households
Water and wastewater	Ratio of households with wastewater collection to total households
Per capita income	Income per person
Electricity access	Ratio of population with electric power and refrigerator (or freezer) to total population
Urban population	Ratio of urban population to total population
Total population	Total number of residents
Pop. under 18	Ratio of population under 18 to total population
Pop. over 60	Ratio of population over 60 to total population
Capital city	1 if located in state capitals or the Federal Capital; otherwise, 0
Metro area	1 for municipalities in a metropolitan area; otherwise, 0
HDI Income	Human development index (Income component)
Income (Poor)	Average income per person among the poor
Income (Extremely poor)	Average income per person among the extremely poor

The variables in the following are used to standardize the data and ensure that we compare similar municipalities (see Table 5).

The estimator (Callaway and Sant'Anna, 2021) is the method used to estimate the average effect of treatment on the treated (ATT) where treatments are staggered over time.

The fundamental assumption in difference-in-differences (DiD) designs is the parallel trends assumption.

Both treated and control municipalities share similar baseline characteristics, such as income levels, urbanization rates, and access to public services, supporting the expectation of parallel morbidity trends before the intervention.

The parallel trends assumption can be expressed mathematically as:

$$E[Y_t(0) - Y_{t-1}(0) \mid D = 1] = E[Y_t(0) - Y_{t-1}(0) \mid D = 0]$$

This assumption posits that the change (or "trajectory") in outcomes over time that units in the treated group would have undergone if they had not been subjected to treatment is identical to the trajectory of outcomes observed among units in the untreated group. This assumption allows for differences in the level of untreated potential outcomes between the groups. It aligns with fixed effects models for untreated potential outcomes, wherein the mean of the unobserved fixed effect may vary across groups.

The Average Treatment Effect on the Treated (ATT) is the difference between the outcome for a group receiving treatment at a particular time and the outcome for the same group if they had not received treatment. The estimator compares the treated group's outcome change between the time they receive treatment and the period just before that. This result is compared to a control group that has not yet received treatment. By comparing these changes, we can estimate the effect of the treatment.

To estimate ATT, we subtract the average outcome change for the control group from the average outcome change for the treated group using sample averages. The estimator looks for identification and estimation under generalizations of the parallel trends assumption to the staggered setting.

Recall that $D_{it} = 1$ if a unit i has been treated by time t, and $D_{it} = 0$ otherwise. Then, for $t = 1, \dots, T-1$, $D_{it} = 1 \implies D_{i(t+1)} = 1$.

Staggered adoption of treatment implies that once a unit participates in the treatment, they remain treated. In other words, units do not "forget" about their treatment experience. Within the DiD context, it is hard to analyze non-staggered treatment setups without further restricting treatment effect heterogeneity across time, groups and treatment sequences.

Parallel Trends Assumption Based on Units Never Transcribed

For all $g = 2, \dots, T$, $t = 2, \dots, T$ with $t \geq g$,

$$E[Y_t(0) - Y_{t-1}(0)|G = g] = E[Y_t(0) - Y_{t-1}(0)|C = 1]$$

This equation denotes the expected difference between the results of a treated unit i at time t ($Y_t(0)$) and time $t-1$ ($Y_{t-1}(0)$), since the unit belongs to the group g. Here, $Y_t(0)$ represents the outcome of unit i at time t in the absence of treatment, and $Y_{t-1}(0)$ represents the outcome at time $t-1$ in the absence of treatment.

In summary, both expressions represent the expected change in outcomes over time for treated units under different conditions: one condition belonging to the treatment group ($G = g$), and the other not being treated ($C = 1$). These expressions are often used in causal inference methods, such as difference-in-difference analysis, to assess the impact of treatments or interventions.

Identification of Group-Time Average Treatment Effects

When we impose the assumption of parallel trends based on 'never treated units', we see that, for all $t \geq g$,

$$\text{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1}|G = g] - \mathbb{E}[Y_t - Y_{g-1}|C = 1].$$

Finally, to compare the result of the single treatment in the state of Tocantins, we employed the simpler version of difference in differences similar to what was previously refined by Saiani and de Azevedo (2018):

$$Y_{it} = \beta_0 + \beta_1 \text{Treated} + \beta_2 \text{Time} + \beta_3 (\text{Treated*Time})_{it} + \mu_i + \epsilon_{it}$$

where Y_{it} is the result of the number of disease cases among a specific age group, β_0 is the intercept, β_1Treated indicates whether or not a municipality was in the treatment group or not, β_2Time whether or not the observation is set before or after privatization, $\beta_3(\text{Treated*Time})_{it}$ is the interaction between time and treatment, μ_i is the fixed effect of the state variable and ϵ_{it} is the error.

The diseases groups are diarrheal diseases, other diseases related to sanitation, and other diseases. The age groups are less than 1 year, 1–5 years, 5–15 years, 15–64 years, and more than 64 years.

The study unfolded in two key stages. Firstly, a control group was meticulously selected, consisting of municipalities that did not undergo privatization and were not located in Tocantins or managed by Saneatins, ensuring their comparability with those that did. The "Propensity Score Matching" (PSM) technique utilized the Probit method to estimate probabilities and balance observed characteristics.

The subsequent step applied the Callaway and Sant'Anna (2021) method to evaluate the difference in the difference estimator between different groups in different periods. We then performed another PSM between only Tocantins municipalities that went fully private, switching from private with state rights to an entirely private provider. Finally, we used the same methodology as Saiani and de Azevedo (2018) to estimate the impact of this switch in the state of Tocantins.

Extending our analysis to include data from diseases unrelated to sanitation was crucial. This additional step is essential to understand whether fluctuations in disease incidence within each municipality can be solely attributed to the impact of sanitation. By examining a broader spectrum of diseases, we can more accurately discern the specific factors contributing to the variations in disease rates and distinguish sanitation's unique role in the observed trends.

Finally, in order to understand the changes that occurred in the state of Tocantins, we conducted another psm, shown below. For the

Table 6
Summary of variables by treatment and control groups - 1991.

Variable	Control (1991)	Treatment (1991)
Per capita income	351.08	357.56
Piped water	0.7167	0.7130
Water and Wastewater	0.6804	0.6755
Pop. under 18	0.4118	0.4152
Pop. over 60	0.0672	0.0656
Total Pop.	119,648	78,057
Urban Pop.	0.7119	0.7083
Electricity access	0.8508	0.8447
Capital city	0.0360	0.0504
Metro area	0.3022	0.2590
HDI income	0.5900	0.5935
Income (Poor)	81.61	81.02
Income (Extremely poor)	44.33	44.30

Table 7
Average treatment effect by Cohort - Diarrheal diseases.

ATT	Standard error	95% confidence interval	Age group
−11.57	5.27	[−21.90, −1.24]*	<1
−15.80	8.43	[−31.42, −0.19]*	1 to 5
−5.77	4.24	[−14.08, 2.54]	5 to 15
−12.35	8.95	[−29.88, 5.19]	15 to 64
−3.16	3.01	[−9.05, 2.74]	>64

Note: Indicates statistical significance at the 5% level.

case of Tocantins, we used a different approach. This had to be done because the smaller number of municipalities resulted in a lack of good exact fits. For this reason, we performed a replacement PSM, having a single municipality matched to several others. We have used weights generated to perform the difference in differences regression.

The Lasso regression model with probit link function was used. The objective is to estimate the coefficients β by minimizing the following objective function:

$$\min_{\beta}\left(\frac{1}{n}\sum_{i=1}^{n} L(y_i, X_i\beta) + \lambda\sum_{j=1}^{p}|\beta_j|\right)$$

- $L(y_i, X_i\beta)$ is the negative log-likelihood function of the probit model for the ith observation.
- λ is the regularization parameter that controls the strength of the penalty term.
- $|\beta_j|$ denotes the absolute value of the coefficient β_j.
- The first term represents the negative log-likelihood loss, which measures the deviation between the observed and predicted responses using the probit link function.
- The second term is the L1 regularization penalty, which encourages sparsity in the coefficient estimates.

6. Results

In this section, we present the results using the difference in difference estimator for multiple periods that range from 1998 to 2021 (see Table 6).

Table 7 statistically significant improvements in health outcomes for children under one year of age and those up to five years of age. These results align well with our expectations, as these age groups are the most vulnerable to poor sanitation conditions.

Although the average treatment effect on treated (ATT) estimator indicates a consistent decline across all age groups, the reductions are significant only for infants and young children. This finding suggests that improvements in sanitation services have the most immediate and pronounced impact on recently born children. These results provide strong evidence that private ownership of sanitation services positively influences health indicators, particularly in reducing morbidity rates among the youngest and most susceptible populations.

Table 8
Average treatment effect by Cohort - Non-Diarrheal diseases.

ATT	Standard error	95% confidence interval	Age group
−0.0038	0.568	[−1.1171, 1.1095]	<1
0.4409	0.7663	[−1.0609, 1.9428]	1 to 5
−0.0653	1.9043	[−3.7977, 3.6671]	5 to 15
2.85	5.8511	[−8.6179, 14.318]	15 to 64
1.2407	1.2017	[−1.1145, 3.5959]	>64

Note: * indicates statistical significance at the 5% level.

Table 9
Average treatment effect by Cohort - Not linked to sanitation.

ATT	Standard error	95% confidence interval	Age group
1.3054	0.4252	[0.4721, 2.1387]*	<1
1.1474	0.6766	[−0.1787, 2.4734]	1 to 5
0.5307	0.3337	[−0.1233, 1.1847]	5 to 15
3.3725	6.297	[−8.9695, 15.7144]	15 to 64
4.4497	2.1747	[0.1873, 8.712]*	>64

Note: Indicates statistical significance at the 5% level.

However, Table 8 shows no observable impact on non-diarrheal diseases. The results show neither improvement nor deterioration in any age group, even after applying multiple propensity score matching techniques. This lack of significant effect was unexpected and highlights a potential fragility in our findings. It suggests that while improvements in sanitation services effectively reduce diarrheal diseases, their influence on other health outcomes remains unclear. This limitation underscores the need for further research to understand the broader health implications of privatized sanitation services.

The results for diseases unrelated to sanitation services in Table 9 show different results compared to the previous tables, with positive and statistically significant effects observed in two age groups. This divergence increases the likelihood that the improvements observed in sanitation-related diseases are not due to random chance but are genuinely attributable to improvements in sanitation services. These findings effectively assess potential bias in the data; if the observed improvements were merely coincidental or driven by external factors, we would expect to see similar trends in non-sanitation-related diseases. The distinct patterns between the two sets of results provide greater confidence in the validity of our conclusions.

The findings from the difference-in-difference estimator applied across multiple periods from 1998 to 2021, indicate a consistent decline in the average treatment effect on health indicators across various age groups, with the most pronounced impact observed in children under five. This decline was statistically significant only for diarrheal diseases in this age group, suggesting that improvements in sanitation services primarily benefited young children and supporting the idea that private ownership of sanitation services had a positive effect on health indicators. In contrast, no significant impact was detected on non-diarrheal diseases. Despite utilizing multiple propensity score matching techniques, no age group exhibited a notable improvement or deterioration in these conditions.

Furthermore, statistically significant positive results for two age groups in sanitation-unrelated diseases further suggest that these effects were not due to chance but rather directly linked to enhanced sanitation services. Overall, these findings highlight the essential role of sanitation services in improving public health outcomes, particularly for vulnerable populations such as infants and young children.

6.1. Tocantins results

Table 11 presents the impact of a post-2014 treatment on different age groups in Tocantins, using census data from 1991. The treatment appears to have significant effects across various age brackets, with a particularly notable decline in the population under five. The difference-in-differences (DID) analysis further highlights substantial

Table 10
Means after matching for Tocantins municipalities - 1991 census.

Variables	1991 control	1991 treatment
Per capita income	193.9676	220.2837
Piped water	0.2124	0.2685
Water and Wastewater	0.1757	0.2220
Pop. under 18	0.5033	0.4956
Pop. over 60	0.0516	0.0588
Total Pop.	21,543	12,794
Urban Pop.	0.4599	0.5330
Electricity access	0.4491	0.5015
Capital city	0.0062	0.0217
Metro area	0.0444	0.0000
HDI income	0.4974	0.5116
Income (Poor)	67.6266	69.7913
Income (Extremely poor)	40.5610	42.5722

Table 11
Tocantins after 2014 (1991 Census).

	<1	1 to 5	5 to 15	15 to 64	>65
Treated	**31.241*****	**58.662*****	**33.458*****	**66.945*****	**16.494*****
	(3.084)	(5.462)	(3.999)	(11.170)	(2.728)
Time	**−60.632*****	**−79.424*****	**−34.718*****	**−135.102*****	**−23.920*****
	(0.723)	(1.280)	(0.937)	(2.618)	(0.639)
DID	**−28.872*****	**−36.821*****	−9.962	10.047	−7.143*
	(4.837)	(8.567)	(6.273)	(17.521)	(4.279)
Controls	**−29.075*****	**−36.918*****	−10.259*	9.039	−7.154*
	(4.809)	(8.453)	(6.172)	(17.271)	(4.169)
Observations	48,478	48,478	48,478	48,478	48,478
R^2	0.218	0.207	0.213	0.163	0.186
Adjusted R^2	0.218	0.207	0.212	0.162	0.185

Note: Indicates statistical significance at the 5% level, Indicates statistical significance at the 1% level, Indicates statistical significance at the 0.1% level.

effects on young children, reinforcing the link between privatization and its impact on the youngest demographic. Notably, even after accounting for additional controls, the magnitude of these effects remains largely unchanged, indicating robustness. However, slight variations between census years suggest potential shifts in demographic patterns over time (see Table 10).

Overall, these findings offer insight into morbidity trends and demographic shifts in Tocantins, carrying important implications for policy and future research. They further reinforce the potential link between private sanitation and improved health outcomes.

7. Conclusion

This investigation presents compelling evidence that challenges the commonly held notion of a trade-off between cost and quality in privately provided public sanitation services. The general results on disease rates are mixed. Although there is an apparent reduction in diseases among children in some cases, not all observations reach statistical significance. In addition, a closer examination of the microdata in the Appendix uncovers statistically significant increases in morbidity within specific subsets of privatized sanitation companies, such as the 2012 cohort. These findings underscore the critical role of local regulation, as highlighted by Carvalho et al. (2023). In contrast, the results from the state of Tocantins provide robust evidence and present a more consistent narrative. In this region, private ownership appears to have been particularly mainly effective, demonstrating the potential for privatization to produce positive outcomes under the right conditions.

These results are consistent with (Galiani et al., 2005), who observed decreased child mortality following water service privatization in Argentina. Similarly, Deutschmann et al. (2023) reported that the privatization of wastewater treatment in Senegal led to a reduction in child diarrhea rates. In contrast, Granados and Sánchez (2014) found slower reductions in child mortality in Colombian municipalities with private sanitation providers, highlighting the importance of the local context and regulatory frameworks. These comparisons underscore the need to consider the specific conditions under which privatization occurs, as they are crucial in determining public health outcomes.

There does not appear to be a clear connection between the privatization of sanitation services and increased access to those services in Brazil. This finding indicates that private operators do not create more infrastructure than public providers. The similarity in infrastructure development may help explain the decrease in morbidity rates in areas where privatization has occurred. Research on access suggests that private operators tend to invest more heavily in expanding wastewater services, whereas public providers focus more on expanding water coverage (Menezes et al., 2016). As a result, the privatization of sanitation services in Brazil seems to lead to improvements in quality, even though neither sector has extensive expansion strategies.

Privately owned sanitation companies appear to have better results in reducing morbidity rates. We can especially see this in the appendix section, where we have more detailed information. The group year 2001 emerged as the most relevant in our findings, where two relevant municipalities with medium to large populations experienced significant reductions in child morbidity.

Some observations regarding the limitations of the study:

- The research utilized data interpolation to fill in gaps from certain months with missing records for specific diseases. While this is a clever way to address incomplete data, it might introduce some biases that could affect the overall analysis.
- Not all sanitation-related diseases were included in the DATASUS database, so the analysis was limited to the diseases that had available data.
- Interestingly, the analysis revealed no significant effects for non-diarrheal diseases, which aligns with trends observed in Tocantins and throughout the country.
- A key finding is that not every cohort group saw improvements, some experienced statistically significant increases in disease incidence. This unexpected result highlights the need for further exploration to understand these dynamics.
- Lastly, identifying a hybrid model of private organization with state rights in just one state suggests a unique context. Exploring various private provision models could yield valuable insights for future research.

These observations pave the way for a deeper understanding of the complexities involved.

Further investigation of concession contracts in the municipalities of Campo Grande (MS) and Sorriso (MS) could improve the policymaker's clarity when choosing the better design of their concession or privatization contracts, leading to substantial decreases in childhood morbidity throughout the country. Both municipalities had their water and wastewater private for an extended period, and the most significant results regarding reducing morbidity rates among children, which are the most sensitive group to bad sanitation services. Therefore, they should be closely monitored to gain more information about what was done.

Another interesting takeaway from the study is the case of Tocantins, where the evidence is very robust and shows a significant trend after 2014 when the company no longer had any interference from the local government. Therefore, we collect clear signs of the superiority of full privatization over private operations with state rights.

Considering the limited resources and fiscal constraints many developing countries face, privatizing public services could offer a viable solution to address the challenges of public health and infrastructure. This research provides valuable information for policymakers to examine successful privatization cases, particularly those significantly reducing childhood diseases.

CRediT authorship contribution statement

Rodrigo França Chaves: Writing – original draft, Software, Data curation. **Adriano Borges Costa:** Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rodrigo Franca Chaves reports administrative support was provided by Insper. Adriano Borges Costa reports a relationship with Insper that includes:. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jup.2025.101903.

Data availability

Data will be made available on request.

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